GE Medical Systems

Technical Publications

Direction 2100618

Revision 1

Signa© Advantage™ 1.0T Quadrature Cervical Spine Coil

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “damage in shipment” written on all copies of the freight or express bill before delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, nature, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785-5052/8*323-5052 immediately after damage is found. At this time be ready to describe name of carrier, delivery date, consignee name, freight or express bill number, item damaged, and extent of damage.

Complete instructions regarding claim procedures are found in section “S” of the Policy & Procedure Bulletins.

3/12/92

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REV 0	11/15/93	Initial Release
REV 1	6/18/96	Add 60cm values to Table 2-2.

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B	Blank						
i	0						
ii	Blank						
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2-1	1						
2-2 — 2-4	0						
3-1 — 3-7	0						
3-8	Blank						
4-1 — 4-4	0						
5-1	0						
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Data1	0						
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GE Logo Pg	-						

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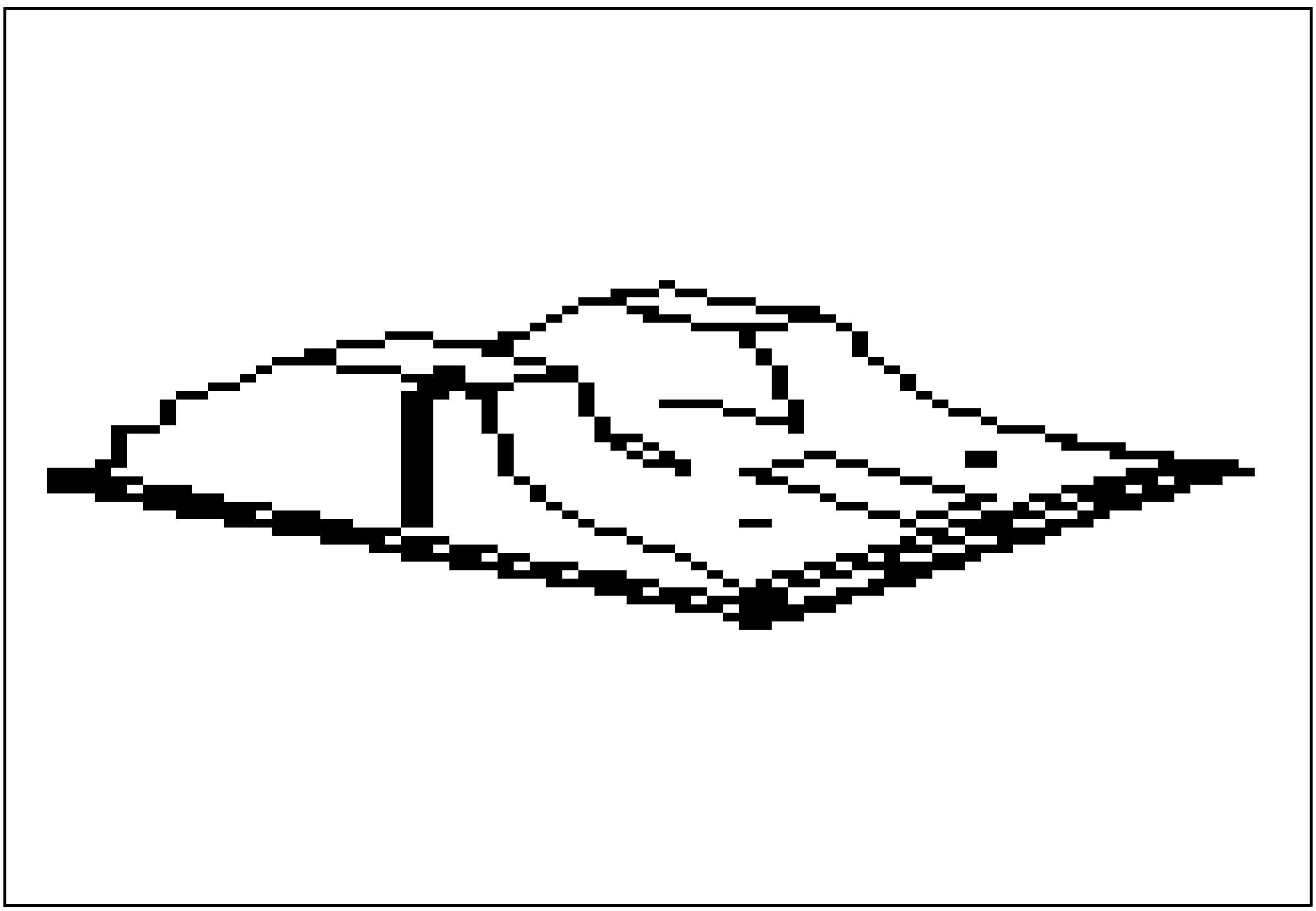
SECTION 1 - INTRODUCTION

1-1 THE QUAD C-SPINE SURFACE COIL

The Quad C-Spine Surface Coil is designed to optimize cervical spine studies and is tuned to that anatomical region. The ROI is 25 cm along the Z axis, 15 cm along the X axis and has an effective depth of at least 12 cm in the Y axis. Two comfort pads are provided to aid in patient comfort. These pads are washable and reusable and are designed to be held in place with Velcro© material that is applied to the back of the pads and on the surface of the coil.

As an aid in operator anatomical landmarking, the coil horizontal and vertical centerlines along with the coil superior/ inferior effective coverage length lines are clearly marked with black tape to provide both visual and tactile information.

The bottom of the coil is supplied with anti-skid pads to prevent coil movement once the coil has been positioned on the patient table. See ILLUSTRATION 1-1.



QUAD C-SPINE COIL
ILLUSTRATION 1-1

1-3 COMPATIBILITY

The Quad C-spine Surface Coil is compatible with the following hardware configurations:

- GE SIGNA ADVANTAGE™ 1.0T Systems

1-4 RELATED DOCUMENTS

- Direction 15400 - Signa Advantage 1.5T, 1.0T, and 0.5T Systems
- Direction 15404 - Signa Advantage 1.5T, 1.0T, and 0.5T Renewal Parts.

1-5 ORGANIZATION OF THIS DOCUMENT

This document is divided into the following sections:

Section 1, INTRODUCTION, describes what the Quad C-spine Surface Coil is and where the Quad C-spine Surface Coil can be used.

Section 2, SET UP AND CALIBRATION, describes installation procedures.

Section 3, FUNCTIONAL CHECKS, describes normal power-up sequence.

Section 4, REPLACEMENT/MAINTENANCE, describes field maintenance procedures

Section 5, RENEWAL PARTS, lists field replaceable service kits and parts.

SNR Data Table

Coil Return Form

TLT Table

1-6 ENVIRONMENTAL REQUIREMENTS

Operate and store the Quad C-spine Surface Coil in the Scanner Room.

SECTION 2 - SET UP AND CALIBRATION

2-1 CHECKING THE SHIPPING LIST

Table 2-1 lists the M1885AM Signa Quad C-spine Coil parts. Check that all parts have been shipped.

TABLE 2-1
COIL PARTS

QTY.	ITEM	PART NO.
1	Quad C-spine Coil	46-328292P1
1	Forehead Strap	N/A
1	Chin Strap	N/A
1	Comfort Pad - Large	N/A
1	Comfort Pad - Small	N/A
1	Service Manual	2100618

2-2 INSTALLING THE QUAD C-SPINE COIL

At the console, install a new soft key. This key will be used by the operator to select Quad C-spine imaging. Name the key "**QUADCSP**".

Refer to the Signa Advantage 1.5T, 1.0T, & 0.5T System Manual for information on installing soft keys (use Coil default values in Table 2-2)

TABLE 2-2
COIL VALUES

SYSTEM	COIL NAME	COIL TYPE	EXTREMITY COIL	CABLE LOSS	COIL LOSS	RECON SCALE FACTOR	XMIT COIL TYPE	XMIT ATTN	MULTI-COIL.
1.0T	QUADC-SP	surface	no	1.24	0.103 - 55cm 0.549 - 60cm	Head Coil Recon Scale Fctr. x 1.00	Quad	0	no

2-3 DETERMINING SYSTEM POLARITY

Before any quadrature coil can be used on the Signa Advantage™ 1.0T System, the system polarity must be determined and the user coil adapted to reflect the same polarity.

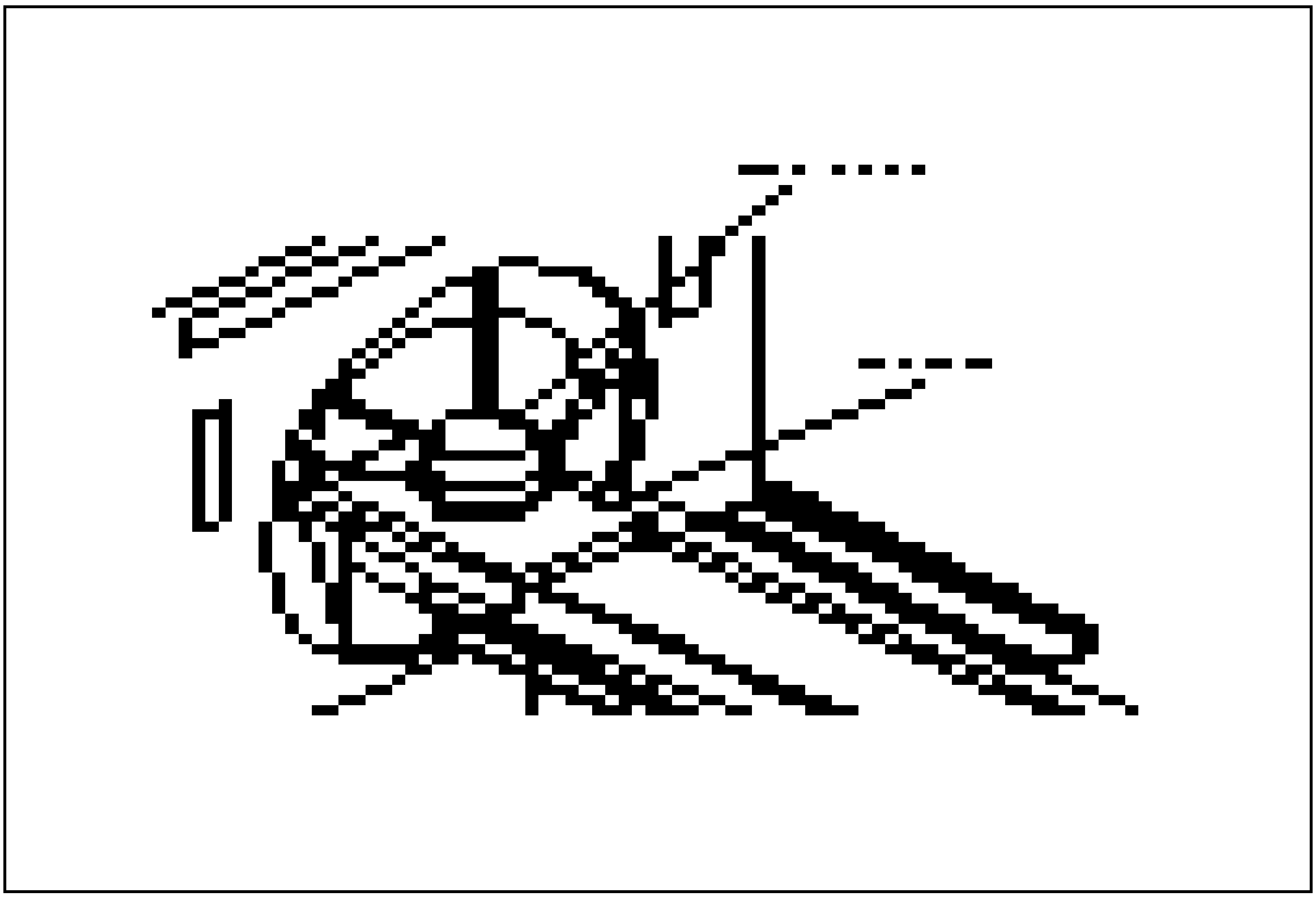
1. Select [**New Exam**] to enable the patient table **LANDMARK** function. Locate the 100 mm phantom on the patient table, see ILLUSTRATION 2-1.

2. Center the 100 mm TLT phantom; X, Y, Z using the positioning lights, **LANDMARK** and advance to isocenter. Set up a scan according to the parameters shown in TABLE 2-3. At the **PREVIEW SCREEN**, press **Scan Ops**. **DO NOT SCAN AT THIS TIME.**

3. Press [**Modify Cvs**]. Type **cfxfull** in the column under **CV Name**.

4. Observe the Current Value column across from cfxfull. If there is a (-) sign in front of the number, the system polarity is **NEGATIVE**. If no sign appears in front of the number, the system polarity is **POSITIVE**.

5. Press [**Enter**] to default to the current value.



PHANTOM PLACEMENT

ILLUSTRATION 2-1

2-4 CHECKING COIL POLARITY

Ensure that the SMA connector is connected to the same polarity fitting as the system polarity. If not, remove the access cover shown in ILLUSTRATION 2-2 by removing the screws. Remove the SMA connector from the fitting on the balun board, and connect it to the other fitting.

CHANGING COIL POLARITY

ILLUSTRATION 2-2

2-5 FUNCTIONAL CHECKS

1. Perform a Body coil scanner performance verification. Refer to Section 3-1, SIGNA ADVANTAGE 1.0T SCANNER VERIFICATION.

2. Perform a Quad C-spine Coil performance verification. Refer to Section 3-2, QUAD C-SPINE COIL IMAGING PERFORMANCE VERIFICATION.

3. Perform an SNR Image analysis. Refer to Section 3-3, SNR IMAGE ANALYSIS**.**

2-6 PERIODIC QUALITY ASSURANCE CHECK

On a periodic basis, such as during planned maintenance, perform the quality assurance check outlined below to ensure that the Quad C-spine coil is operating properly with no appreciable degredation of image quality.

1. Check the external cable for cracks or breaks once each week. See Section 4-5, CHECKING THE EXTERNAL CABLE.

2. Perform a Quad C-spine Coil performance verification. Refer to Section 3-2, QUAD C-SPINE COIL IMAGING PERFORMANCE VERIFICATION.

3. Record the date and value calculated in SNR QA DATA TABLE

4. As instructed in the Data Table, divide the SNR value obtained in the periodic QA check by the original SNR value.

5. If this ratio is not greater than 85%, then there may be a problem with the Quad C-spine Coil or the MR system. Contact the GE Service Representative.

TABLE 2-3
SCAN PARAMETERS

ID:	**geservice**	Auto CF:	**[Peak]**
Name:		Receive Bandwidth:	**[16kHz]**
Weight (lb):	**111**	Field of View:	**[24 cm]**
Patient Entry:	**[Head First]**	Slice Thickness:	**[10 mm]**
Patient Position:	**[Supine]**	Interscan Spacing:	**[0]**
Axial/Sag Landmark:	**[Sternal Notch]**	Start Location (I/S):	**0**
Coil Type:	**[++] See Note**	End Location (I/S):	**0**
Scan Plane:	**[Axial]**	No of Scan Locations:	**1**
Image mode:	**[2D]**	FOV Center (L/R):	**0(P/A):0**
	[Monitor SAR]	Acq. Matrix: (freq.)	**[256]**
Pulse Sequence:	**[Spin Echo]**	Acq. Matrix: (phase)	**[128]**
Imaging Options:	**[None]**	Frequency Direction:	**[R/L]**
Number of Echos:	**[*] See Note**	Phase FOV:	**default**
Echo Time (TE):	**[20 ms]**	Imaging Time:	**[1 Nex]**
Rep Time (TR):	**[Other] 600 ms**	Contrast:	**[No]**
		Table Delta	**0**

NOTE: * **[4]** for Section 3-1-2; 3-2-2
[1] for Section 3-1-4; 3-2-4
++ **[Body]** for Section 3-1-2; 3-2-2
[Other][QUADCSP] for Section 3-1-4; 3-2-4

SECTION 3 - FUNCTIONAL CHECKS

3-1 SIGNA ADVANTAGE™ 1.0T SCANNER VERIFICATION

Note

An alternative proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located in Direction 15491, *Signa Advantage 1.5T, 1.0T, and 0.5T System Troubleshooting Guide.*

1. Scanner performance in the body coil mode should be verified. The RF power level calibration from the body coil mode is used to verify the surface coil decoupling. Landmark the 100 mm TLT phantom without the coil. Place the TLT Phantom on the Universal Phantom Holder. See ILLUSTRATION 2-1.

2. Set up the scan using the parameters in TABLE 3-1. Coil Type: [**Body Coil**] and Number of Echos: [**4**] should be inserted. Perform an **Auto Prescan** to calibrate the RF power level. Record the value for TG, R1, and R2.

3. Run the scan. Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images.

4. Select [**Cancel**], [**New Series**], to perform SNR comparisons. Acquire two identical images using the previously loaded scan parameters in TABLE 3-1after inserting Number of Echos: [**1**]. Immediately after acquiring the

TABLE 3-1
SCAN PARAMETERS

ID:	**geservice**	Auto CF:	**[Peak]**
Name:		Receive Bandwidth:	**[16kHz]**
Weight (lb):	**111**	Field of View:	**[24 cm]**
Patient Entry:	**[Head First]**	Slice Thickness:	**[10 mm]**
Patient Position:	**[Supine]**	Interscan Spacing:	**[0]**
Axial/Sag Landmark:	**[Sternal Notch]**	Start Location (I/S):	**0**
Coil Type:	**[++] See Note**	End Location (I/S):	**0**
Scan Plane:	**[Axial]**	No of Scan Locations:	**1**
Image mode:	**[2D]**	FOV Center (L/R):	**0 (P/A): 0**
	[Monitor SAR]	Acq. Matrix: (freq.)	**[256]**
Pulse Sequence:	**[Spin Echo]**	Acq. Matrix: (phase)	**[128]**
Imaging Options:	**[None]**	Frequency Direction:	**[R/L]**
Number of Echos:	**[*] See Note**	Phase FOV:	**default**
Echo Time (TE):	**[20 ms]**	Imaging Time:	**[1 Nex]**
Rep Time (TR):	**[Other] 600 ms**	Contrast:	**[No]**
		Table Delta	**0**

NOTE: * **[4]** for Section 3-1-2; 3-2-2
[1] for Section 3-1-4; 3-2-4
++ **[Body]** for Section 3-1-2; 3-2-2
[Other][QUADCSP] for Section 3-1-4; 3-2-4

first image, select **Scan** to obtain a second. Note the series number for future reference.

3-2 QUAD C-SPINE COIL IMAGING PERFORMANCE VERIFICATION

1. Set up the Quad C-spine coil and 100 mm phantom as shown in ILLUSTRATION 3-1. Connect the surface coil cable to the **SURFACE COIL QUICK DISCONNECT BOX** and connect to the **HEAD COIL PORT.**

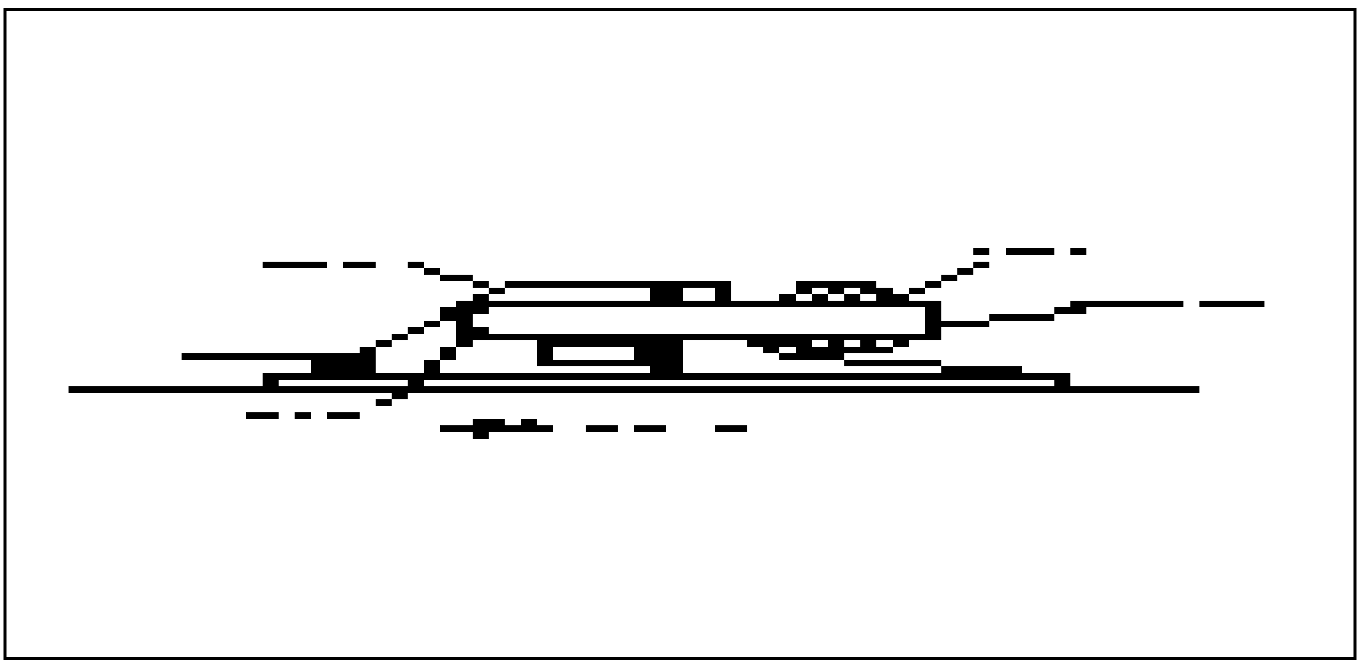
2. Select [**Cancel**], [**New Series**], and repeat the scan using the parameters in TABLE 3-1. Coil Type: [**Other**][**QUADCSP**] and Number of Echos: [**4**] should be used. Use **Auto Prescan** to set the values of R1 and R2 to those in Section 3. If the value of TG does not match the value obtained in step 3-1-2, use **Manual Prescan** to set the value of TG.

3. Run the scan. Ensure that there are no artifacts of any sort on any of the images. Study the image using a **Window Width** and **Window Level** to obtain good contrast. A properly functioning coil will produce an image with a smooth signal pattern, gradually dropping in intensity as the distance from the coil increases, see ILLUSTRATION 3-2. A decoupling failure will cause holes, distortions, or striped patterns in the image, Refer to Section 3-4 to 3-8 to troubleshoot decoupling failures.

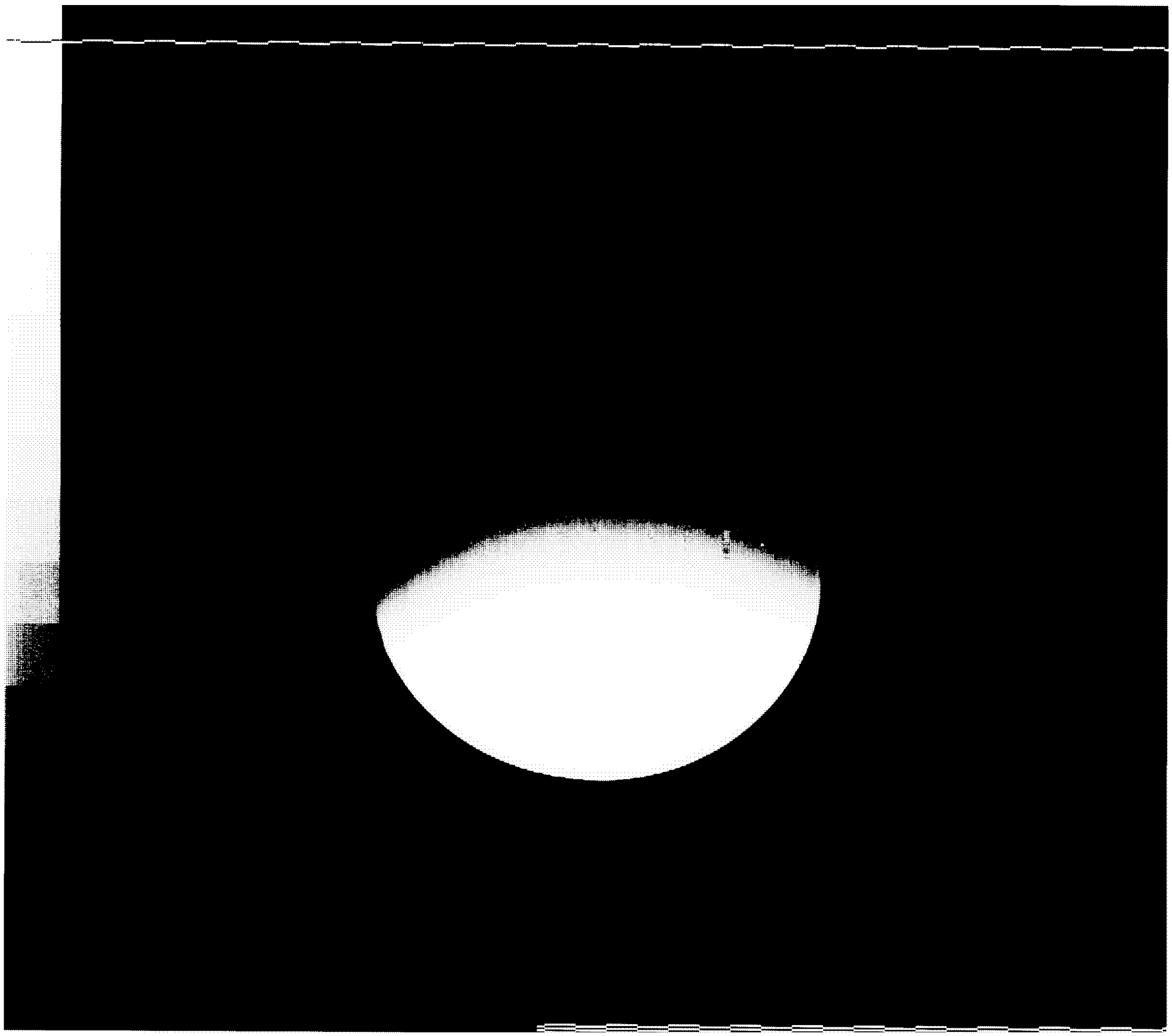
4. Repeat step 2 with Number of Echos: [**1**] inserted. Run the scan. Immediately after acquiring the first image, select **Scan** to obtain a second.

5. Compute the SNR using the procedure described in Section 3-3, SNR IMAGE ANALYSIS. Compare the signal to noise ratio of the first echo of the Quad C-spine Coil images with the signal to noise ratio of the first echo of the body coil images of step 3-1-4. The Quad C-spine Coil scans should display a signal to noise ratio greater than 2.5 times that of the body coil scans.

6. Inspect the images for visible ghosting (similar in appearance to motion artifacts) in the phase encoding direction.



PHANTOM PLACEMENT
ILLUSTRATION 3-1



NORMAL DECOUPLING
ILLUSTRATION 3-2

If images contain ghosting artifacts, suspect an intermittent cable or connection. Refer to Section 3-4 to 3-8 for troubleshooting information.

3-3 SNR IMAGE ANALYSIS

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center for positioning the ROI. A difference image is created by subtracting the second image from the first and to calculate noise from the subtracted image. Signal value, noise value, and SNR are reported. The difference image can be saved with the results annotated (option).

Procedure

1. Touch [**Utilities**], [**MR Tools**], [**Image Quality**], then [**SNR Test**]. The SNR Test screen is displayed. See ILLUSTRATION 3-3.

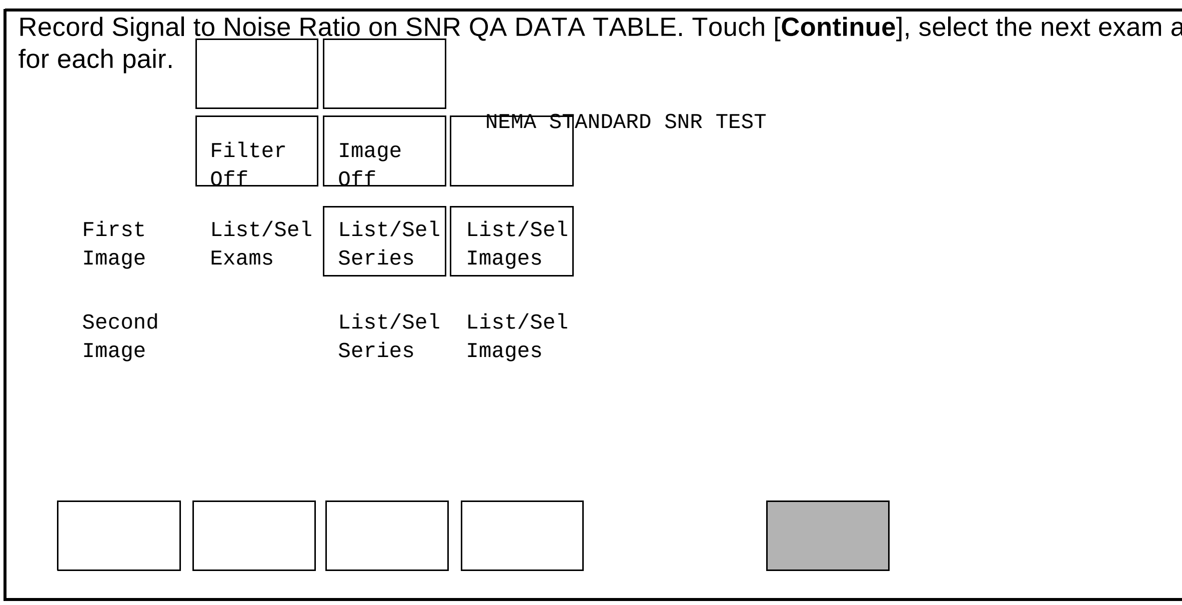
2. Enter first image exam, series, and image numbers. If exam, series, or image numbers are not known, select [**List Exams**], [**List Series**], or [**List Images**] to display list to choose from. Enter second image series and image number. The second image Exam Number is defaulted to the same as first image.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

3. Select [**Filter Off**] and [**Image Off**]. Touch [**Accept**] to begin analysis. The final values are displayed on the screen.

4. Record Signal to Noise Ratio on SNR QA DATA TABLE. Touch [**Continue**], select the next exam and repeat for each pair.



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 3-3

```
Cancel    Utility   MR Tools  Image                     Accept
          Main      Main      Quality
```

3-4 COIL IMPEDANCE MEASUREMENTS

1. Move to an area where the magnetic field from the Signa Advantage 1.0T magnet is less than 5 Gauss. Load the surface coil with a human subject. Use the procedure in the Operation Manual to place the surface coil.

2. Connect the surface coil cable to the vector impedance meter probe using an appropriate adapter. Use the vector impedance meter to measure the coil impedance at 42.68 MHz. Ensure that the cable does not touch or pass close to the coil housing while making this measurement.

3. The impedance magnitude should nominally be 50 ohms under loaded conditions. It should fall within the range of 37.5 to 62.5 ohms under test conditions.

4. The phase angle should nominally be -10 degrees. It should fall within the range of -25 to +5 degrees under test conditions.

3-5 COIL INTEGRITY TESTS

1. Repeat Section 3-4 while flexing the surface coil cable, especially at the fitting and the coil housing strain relief. Using the **HIGH SPEED** mode on the vector impedance meter, note any erratic fluctuations in the coil impedance. Any erratic or unstable readings may indicate a defective cable or connector. Refer to Section 4, REPLACEMENT/MAINTENANCE.

3-6 PROTECTION DIODE TESTS

1. Use the digital multimeter on the diode test range to measure across the coil connector. With the negative lead to the connector shell and the positive to the connector center pin, the reading should be between 0.5 and 1.0. Reversing the leads should cause the reading to be infinity (overrange).

2. An open reading in both directions indicates an open diode, or an open cable. Refer to Section 3-8 to check the cable. Refer to Section 4-3, DIODE REPLACEMENT to replace the diode.

3. A short circuit or excessively low reading indicates a shorted diode or a shorted cable. Refer to Section 3-8 to check the cable. Refer to Section 4-3, DIODE REPLACEMENT to replace the diode.

3-7 DECOUPLING DIODE TESTS

WARNING

DO NOT MOVE THE BALUN BOARD COMPONENTS WHEN MAKING THESE MEASUREMENTS. ANY MOVEMENT OF THE COMPONENTS MAY DETUNE THE COIL CIRCUITRY.

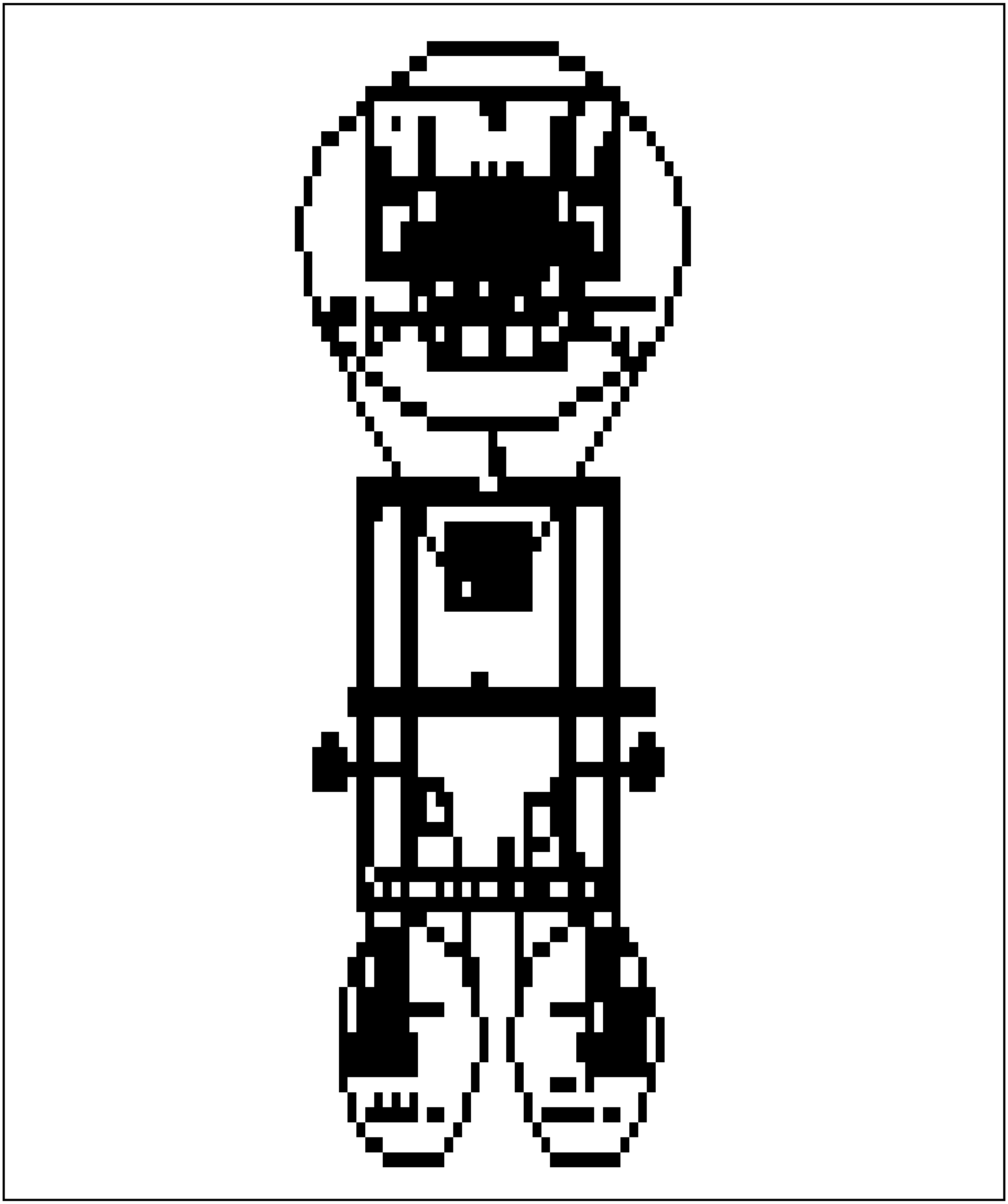
Note

Decoupling diodes are not field replaceable. Coils with decoupling diode failures must be replaced with a new coil assembly.

1. Remove the service access cover. Unscrew the two decoupling diode access plugs. See ILLUSTRATION 3-4.

2. Locate the two pairs of test points for the decoupling diodes labeled **COIL A TEST POINTS** and **COIL B TEST POINTS.**

3. Using the diode test function on the digital multimeter, measure across each pair of points in sequence, first



COIL TEST POINTS
ILLUSTRATION 3-4

across the pair marked **A**. A reading between 0.5 and 1.0 should be obtained. Reverse the leads on the same test points to reverse the polarity of the multimeter. A similar reading should be obtained.

3-7 DECOUPLING DIODE TESTS (continued)

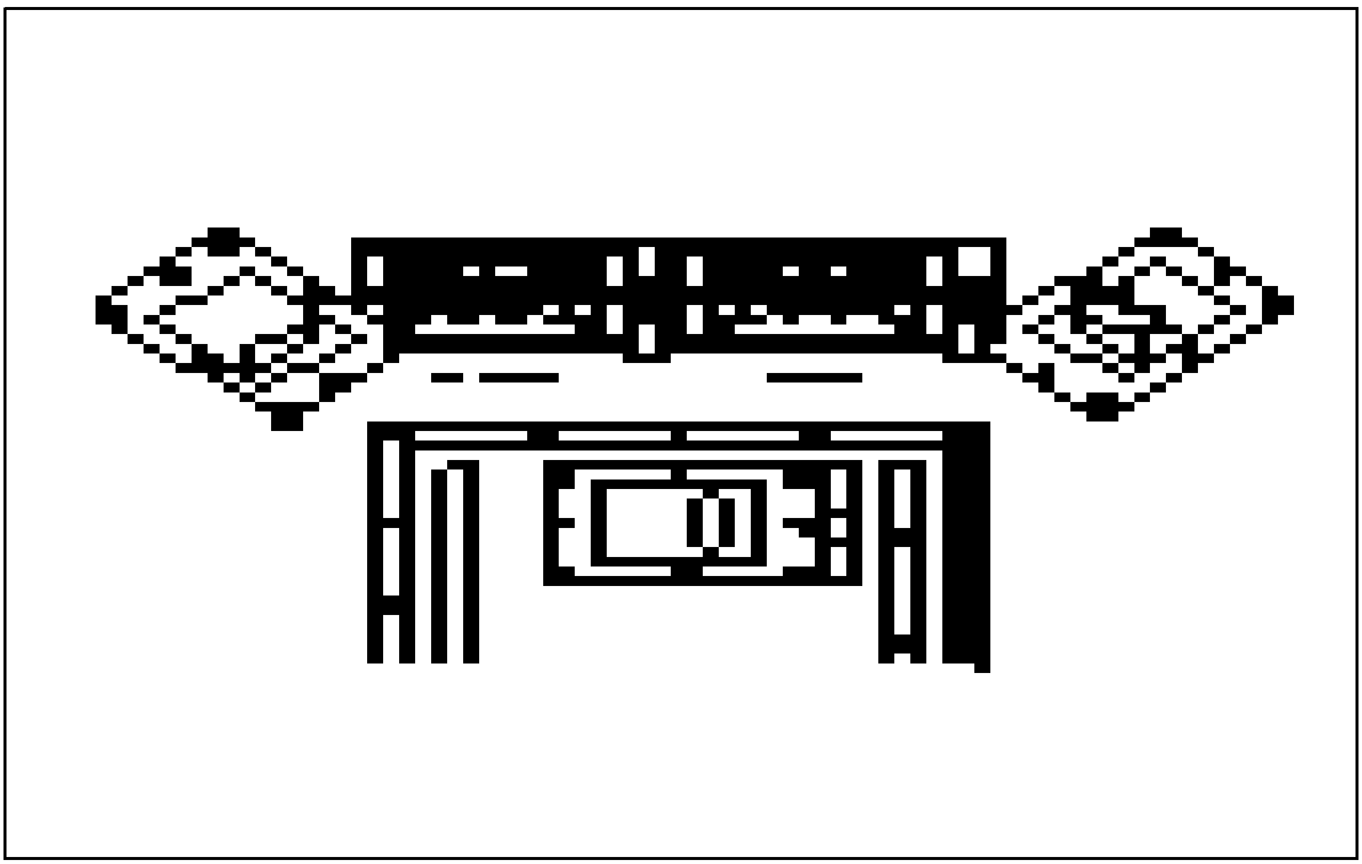
4. An overrange in both directions indicates that both diodes of a pair are open. This condition causes a portion of the decoupling circuitry to be inoperative, and the coil must not be used clinically.

5. A reading near zero indicates one or more shorted diodes. While decoupling is not affected, the signal to noise ratio of the coil will be degraded.

3-8 CABLE TESTS

1. If the surface coil passes the tests in Sections 3-4, 3-5, and 3-6, the cable can be assumed to be functional.

2. If a short or open circuit was discovered in Section 3-4, the cable may be tested by disconnecting the cable at the PC board via the service access cover, as shown in ILLUSTRATION 3-5. The cable may then be tested for continuity of the center conductor and shield, and for shorts between the center conductor and shield using the multimeter continuity test function.



ACCESS COVER, INTERNAL CABLE CONNECTION
ILLUSTRATION 3-5

SECTION 4 - REPLACEMENT/MAINTENANCE

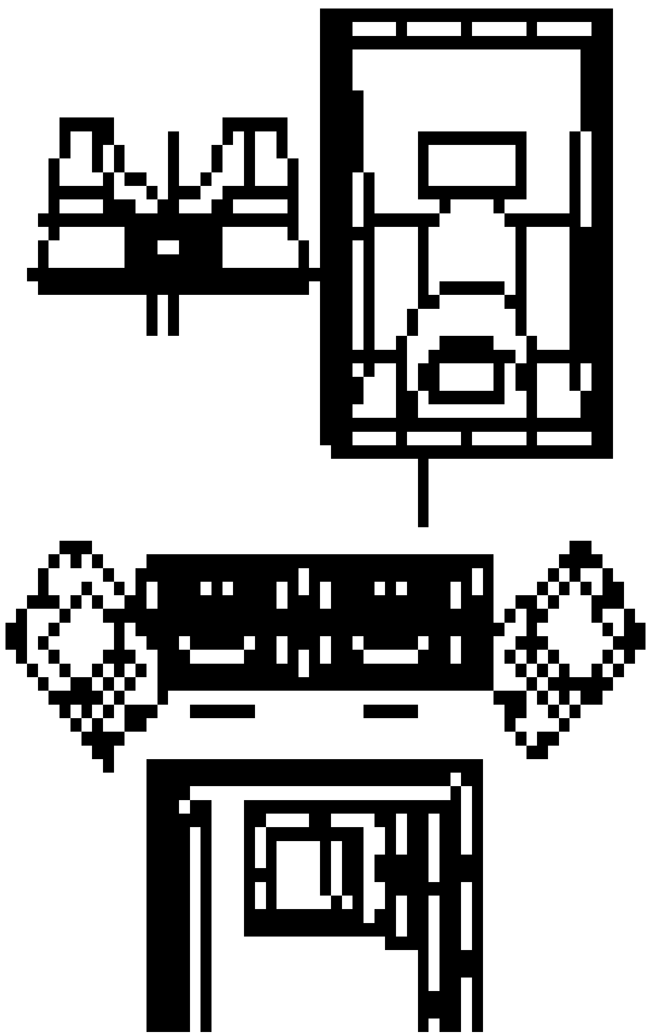
4-1 COAXIAL CABLE REPLACEMENT

1. Remove the service access cover from the housing by removing the eight brass screws attaching the cover. See ILLUSTRATION 4-1

2. Disconnect the cable SMA connector from the PC board SMA connector.

3. Loosen and remove the strain relief body from the strain relief base. Slide the strain relief body down the cable away from the base.

4. Loosen and remove the retaining nut from the strain relief base.



COAXIAL CABLE REPLACEMENT
ILLUSTRATION 4-1

4-1 COAXIAL CABLE REPLACEMENT - (continued)

5. Remove the cable assembly by removing the strain relief base from the hole in the coil housing.

6. Remove the retaining nut from the new cable assembly. Insert the SMA end of the cable through the strain relief mounting hole in the coil bottom cover. Place the retaining nut over the SMA end of the cable.

7. Connect the cable SMA connector to the PC board SMA connector. Tighten firmly.

8. Attach the strain relief base by inserting it into the hole in the coil housing. Install and tighten the retaining nut. Tighten the nut one-half turn beyond initial contact with the surface of the housing.

9. Install and tighten the strain relief body to the base until the base begins to turn (8 inch-pounds).

10. Attach the service access cover to the coil housing.

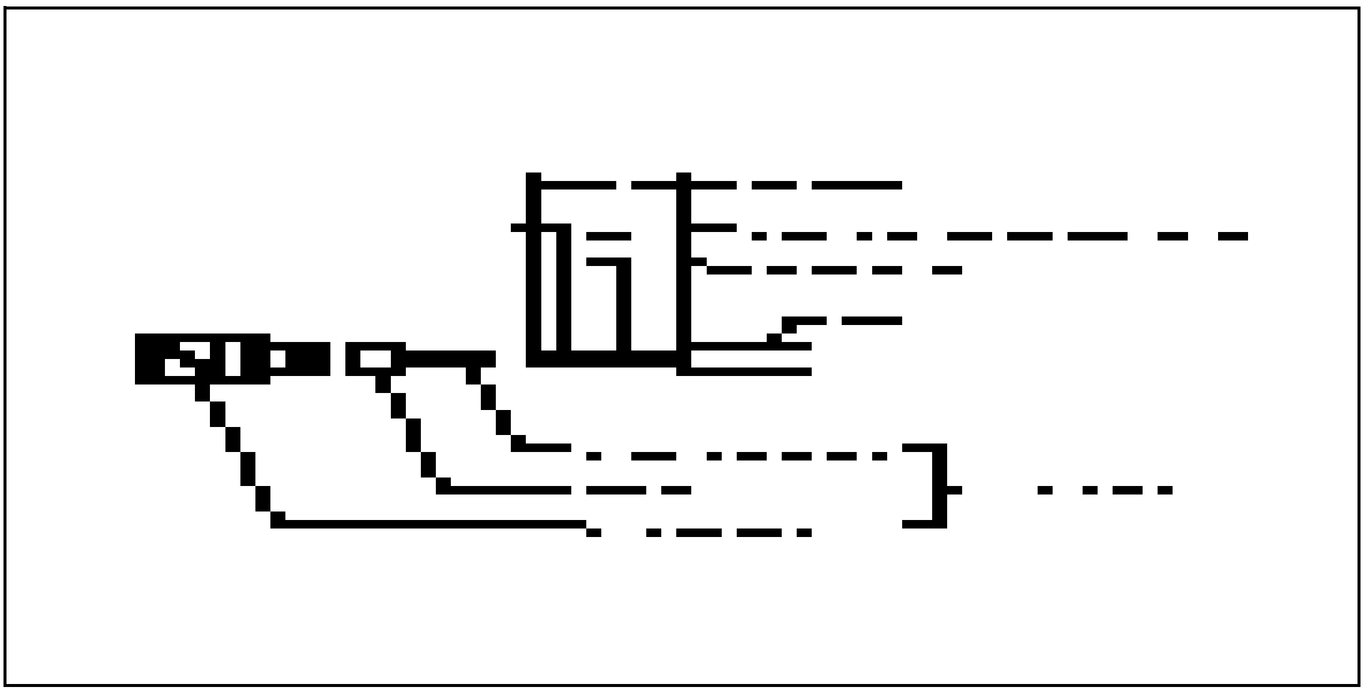
11. Repeat the tests in Sections 3-2 and 3-3 to verify proper coil operation.

4-2 OUTPUT CONNECTOR REPLACEMENT

1. If the cable will be shortened to less than 56-3/4 inches by a repair, it is recommended that the entire cable be replaced in the event of a connnector or connector-end cable failure.

2. Replace the connector using ILLUSTRATION 4-2 as a guide. Use a new BNC coaxial fitting - part number 46-271494P1, and crimp tool - part number 46-255841, with insert - part number 46-255841P100, to install the replacement connector on the cable.

3 Repeat the tests in Sections 3-2 and 3-3 to verify proper operation.



OUTPUT CONNECTOR REPLACEMENT
ILLUSTRATION 4-2

4-3 DECOUPLING DIODE REPLACEMENT

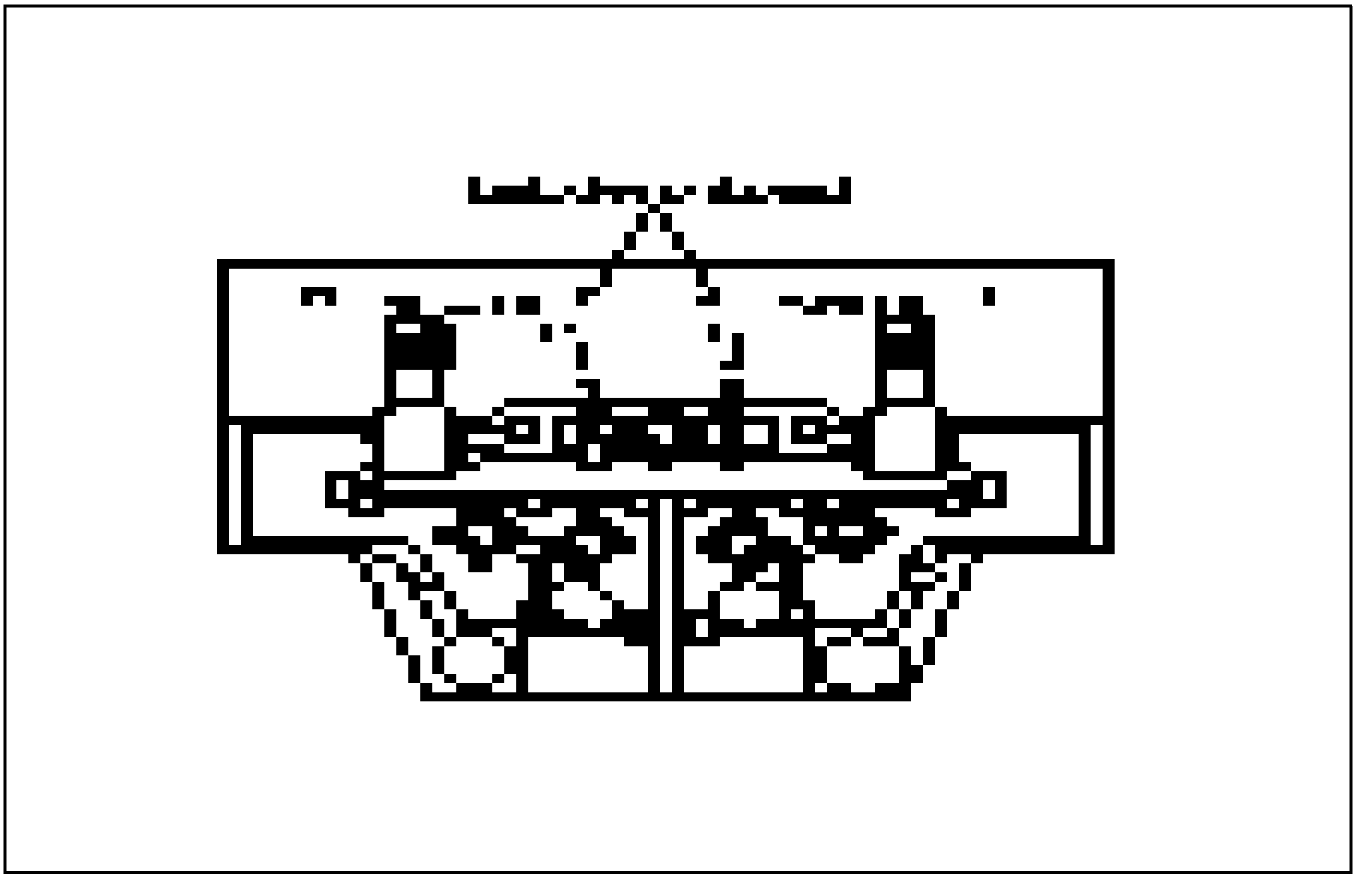
1. As coil tuning orthogonality, and decoupling may be affected by diode replacement, decoupling diodes are not field replaceable. Coils with decoupling failures must be replaced.

4-4 PROTECTION DIODE REPLACEMENT

1. If it has been determined that a diode must be replaced, remove the access cover.

2. Diode placement is shown in ILLUSTRATION 4-3.

3. Unsolder both diodes and replace with part number 46-221735P1. Place the cathodes to ground.



DIODE REPLACEMENT
ILLUSTRATION 4-3

4-5 REPLACING THE MECHANICAL HARDWARE

1. Replace the entire Quad C-Spine Coil. Refer to Section 5, RENEWAL PARTS, for Quad C-Spine Coil part number.

4-6 CHECKING THE EXTERNAL CABLE

1. Check the external cable for cracks or breaks once eack week. Replace the external cable if any damage or wear is found. Refer to Section 4-2, OUTPUT CONNECTOR REPLACEMENT, for replacement procedure.

4-7 CLEANING THE QUAD C-SPINE COIL

1. Clean the Quad C-Spine Coil and external cable with a mild dishwashing liquid and water solution. Wet a soft cloth with the solution and proceed to clean.

SECTION 5 - RENEWAL PARTS

5-1 SERVICE KITS

1. A service kit is available for replacement of the cable assembly:

CABLE KIT 2100937-2 — Consisting of the following items:

ITEM	DESCRIPTION	QTY
1	Cable Assembly	1

2. The Quad C-spine Coil Part number is:

ITEM	DESCRIPTION	QTY
1	Quad C-spine Coil with Cable—46-328292P1	1

Blank

SNR QA DATA TABLE

Use the space provided below to record the calculated signal to noise ratio—SNR, data obtained from Section 4, *SNR IMAGE ANALYSIS*. After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2, enter the date in column 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: ______________________

Quad C-spine SNR value: ______________________

Date: ______________

SNR DATA QA (QUALITY ASSURANCE) CHECK:

1 Date	**2** Quad C-spine Coil QA SNR Value:	**3** Is new value divided by original value >85%?	
______	____________	______________________	**Y/N**
______	____________	______________________	**Y/N**
______	____________	______________________	**Y/N**
______	____________	______________________	**Y/N**
______	____________	______________________	**Y/N**
______	____________	______________________	**Y/N**
______	____________	______________________	**Y/N**
______	____________	______________________	**Y/N**
______	____________	______________________	**Y/N**
______	____________	______________________	**Y/N**
______	____________	______________________	**Y/N**

Make additional copies of this document as needed.

DEFECTIVE SURFACE COIL RETURN FORM

NOTE

For proper assessment of defective returned coils, both sides of this form should be completely filled out and accompany all returned coils. Include films or prints of any image quality complaints with a description of the scan prescription used.

DATE: ______________________________

SITE NAME: ______________________________

SITE ADDRESS: ______________________________
____
____

SERVICE ENGINEER: ______________________________

COIL SERIAL NUMBER: ______________________________

DATE COIL INSTALLED: ______________________________

DESCRIPTION OF COIL PROBLEM: ______________________________
[Please be descriptive;"Broken" is not enough] ______________________________
____

ELECTRICAL CHECKS	
VECTOR IMPEDANCE METER TEST - COIL LOADED WITH HUMAN SUBJECT IN FREE SPACE—Refer to Section 3-4-1.	
MAGNITUDE	
PHASE	
VECTOR IMPEDANCE METER TEST - COIL UNLOADED IN FREE SPACE—Refer to Section 3-4-1.	
MAGNITUDE	
PHASE	
PIN DIODE TEST—Refer to Section 3-4-3.	
DIODE DROP—FORWARD BIAS	
DIODE DROP—REVERSE BIAS	

Note

An alternate proprietary procedure is available to GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located in Direction 15491, *Signa Advantage 1.5T, 1.0T & 0.5T System Troubleshooting Guide.* Use the table to record the TLT results of the defective coil.

TABLE DR-1
TLT DATA FOR DEFECTIVE SURFACE COIL

SITE:	**NAME:**	**DATE:**
TLT FILE NUMBER:		
SNR: NOISE		
SNR MEAN		
SNR SIGNAL		
SNR AREA		
TRMAP: 89–91	%	%
85–95	%	%
65–115	%	%
FLIP MEAN		
FLIP SDV		
RCV MEAN		
RCV SDV		
DIFF MEAN		
DIFF SDV		

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